ARUN BALASUBRAMANIAM

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RESEARCH SKILLS

- **Tissue Culture:** Human primary enteric neuronal cells, mouse primary enteric neuronal cells, murine immorto fetal enteric neuronal cells (IM-FEN), mouse embryonic fibroblast cells (NIH3T3), and human keratinocyte cells (HaCaT).
- **Molecular Biology:** Western blotting, immunofluorescence (IF), immunohistochemistry (IHC), whole-mount immunostaining, RNA, DNA, protein extraction, cDNA synthesis, RT-qPCR, biochemical assays, and ELISA, and AAV-based gene therapy (design, packaging, and delivery).
- **Imaging:** Confocal microscopy, fluorescence microscopy, brightfield microscopy, iDISCO.
- **Animal Work:** Mouse whole body dissection, serum collection, oral gavage, injection techniques (subcutaneous, intravenous, intraperitoneal, retro-orbital, tail-vein), histological sectioning, whole-gut transit, and colon-bead test.
- **Software:** Microsoft Office, ImageJ, R-studio, and GraphPad Prism.

EDUCATION

Ph.D. **2021**
Biomedical Engineering
National Central University
Taiwan.

B.S **2016**
Genetic Engineering
SRM University
India.

RESEARCH EXPERIENCES

Postdoctoral Researcher, Emory University, Atlanta, USA **05/2021 ~ Present**

- Established research on the impact of Western diet-induced ferroptosis and its effect on enteric neurodegeneration and bowel-movement gastrointestinal (GI) disorder.
- Proposed and researched the impact of nano-microplastics on enteric neuronal mitochondria in gastrointestinal disorders.
- Guided undergraduate students to enhance their understanding of basic molecular techniques.

Ph.D. Researcher, National Central University - Taiwan **09/2016 ~ 01/2021**

- Proposed and established microbiome research focused on the role of skin probiotic bacteria in neutralizing free radicals.
- Managed tissue culture laboratory and maintained cell lines, chemical reagents, and protocols.
- Mentored master's students in their academic work and thesis.

RESEARCH GRANTS AND AWARDS

- **2026-2028** - NIH/NIDDK K99/R00 Pathway to Independence Award (1K99DK149050-01), "Mechanism of Nanoplastics Induced Gastrointestinal Neurogenic Inflammation." Principal Investigator; award period 09/2026 - 08/2028.
- **2026** - AGA Research Foundation Fellow Abstract Award, Digestive Disease Week 2026.
- **2025-2026** - Emory Postdoctoral Academic Appreciation Award, Emory University, recognizing first-author publication.
- **2025** - Shanthi Sitaraman Award at Digestive Disease Research Day, jointly organized by Georgia State University, and the Emory Digestive Diseases Initiative.
- **2025** - ANMS basic science Sabbatical grant for a scientific visit to Ohio State University, USA.
- **2022** - Received a Poster of Distinction Award and had abstract ranked in the top 10% at DDW 2022 in San Diego, USA.
- **2022** - Awarded the Division of Digestive Diseases Travel Grant for the academic year.
- **2018-2020** - Received the MOST-Taiwan Academic-Industrial Ph.D. Fellowship.
- **2016-2017** - Awarded the National Central University's M.S. Fellowship.

PUBLICATIONS

Published research articles:

- Western diet induces iron-dependent enteric neurodegeneration via ferroptosis. (Journal of Clinical Investigation; IF-13.6, 2026) **Arun Balasubramaniam**, Dmitrii Pavlov, Yunpeng Du, Jeremy Reeves, Alan Harzman, Yunshan Liu, Francesca Cingolani, Xinxu Yuan, Jay M. Patel, Simon Musyoka Mwangi, Peijian He, C. Michael Hart, Wenhui Hu, Fievos L. Christofi, Shanthi Srinivasan*
- Enteric neuronal vasoactive intestinal peptide mediates SARS-CoV-2 associated diarrhea. **Arun Balasubramaniam**; PR Tedbury; A Gracz; SM Mwangi; G Li; D Merlin; SG Sarafianos; P He; S Srinivasan* (Biomolecules; IF- 6.064; 2022)
- Lipid nanoparticle technology for therapeutic GDNF delivery to the liver. Chunhua Yang, Simon Musyoka Mwangi, **Arun Balasubramaniam**, Ge Li, Olivier Merlin Zhang, Yunshan Liu, Shanthi Srinivasan* (Gastro Hep Advances; 2023)
- SIRT3 activation promotes enteric neurons survival and differentiation. **Arun Balasubramaniam**; Ge Li; Anita Ramanathan; Simon Musyoka Mwangi; Michael C. Hart; Jack L. Arbiser; Shanthi Srinivasan* (Scientific Reports; IF- 4.379; 2022)
- Palmitate and Western diet induce oxidative damage and impairment of mitophagy in hepatocytes - reversal by a neurotrophic factor (Scientific Reports; IF- 4.379; 2022) SM Mwangi; G Li; **Arun Balasubramaniam**; D Merlin; CM Hart; MJ C; Shanthi Srinivasan*
- Repurposing INCI-registered compounds as skin prebiotics for probiotic Staphylococcus epidermidis against UV-B. (Scientific Reports; IF- 3.998; 2020) **Arun Balasubramaniam**; Adi, P.; Do Thi; Keshari, S; S Raman, Chien-Lung Chen; C-M Huang*
- Skin Bacteria Mediate Glycerol Fermentation to Produce Electricity and Resist UV-B (Microorganisms; IF- 4.125; 2020) **Arun Balasubramaniam**; Adi, P.; Do Thi, T.M.; Yang, J.-H.; Labibah, A.S.; Huang, C.-M.
- Butyric Acid from Probiotic Staphylococcus epidermidis in the Skin Microbiome Down-Regulates the Ultraviolet-Induced Pro-Inflammatory IL-6 Cytokine via Short-Chain Fatty Acid Receptor (IJMS; IF- 4.556; 2019) Keshari, S.; **Arun Balasubramaniam**; Myagmar B.; Herr, D.R.; Negari, I.P.; Huang, C.-M*
- Leuconostoc mesenteroides mediates an electrogenic pathway to attenuate the accumulation of abdominal fat mass induced by a high-fat diet. (Sci Rep; IF- 3.998; 2020) Minh Tan Pham, John Jackson Yang, **Arun Balasubramaniam**, Adelia Riezka Rahim, Prakoso Adi, Thi Tra My Do, Deron Raymond Herr, C-M Huang*.
- Colonization of nasal cavities by Staphylococcus epidermidis mitigates SARS-CoV-2 nucleocapsid phosphoprotein-induced interleukin (IL)-6 in the lung. M-S Kao; J-H Yang; **A Balasubramaniam**; S Traisaeng; D R. Herr; C-M Huang (M Biotechnology; IF-6.575; 2022)

Published commentary:

- Diet-Microbiome-ENS connection: Impact of the Cafeteria Diet. **A Balasubramaniam**, S Srinivasan* (AJP-Gastrointestinal and Liver Physiology IF- 3.9; 2025)
- Molecular mechanisms of enteric neuropathies is diabetes and high-fat diet feeding. Cingolani, Francesca, **Arun Balasubramaniam**, S Srinivasan* (NGM; IF- 3.960; 2023)
- Role of Stimulator of Interferon Genes (STING) in the enteric nervous system in health and disease. **Arun Balasubramaniam**; S Srinivasan* (NGM; IF- 3.960; 2023)
- Dysbiosis associated with Esophageal Adenocarcinoma- a novel method to detect tissue-associated microbiome. (GHepAdvance; 2022) **A Balasubramaniam**; S Srinivasan*

Published review articles:

- Prospects of acne vaccines targeting secreted virulence factors of Cutibacterium acnes (Expert Review of Vaccines; IF- 4.370; 2019) S Keshari, M Kumar, **Arun Balasubramaniam**, T-W Chang, Y Tong, C-M Huang*

PEER-REVIEWED JOURNALS

- Nature Microbiology 2022.
- Journal of Molecular Medicine 2021.
- Neurogastroenterology and Motility 2021.
- Scientific Reports -2021.
- JCI Insight-2023
- Journal of Molecular Medicine-2022

- New England Journal of Medicine-2023
- Pediatric Research-2023
- Pharmacological Research-2024
- Medcomm-2024
- Annals of Medicine-2024
- British journal of Pharmacology-2025

SCIENTIFIC SOCIETY MEMBERSHIPS

- Crohn's & Colitis Foundation (CCF), USA (2025-present).
- American Gastroenterological Association (AGA), USA (2025-present).
- American Neurogastroenterology and Motility Society (ANMS), USA (2023-present).
- American Society for Biochemistry and Molecular Biology (ASBMB), USA (2019-2020).
- Society for Free Radical Research Europe (SFRR Europe), Spain (2019-2021).

CONFERENCE POSTER & ORAL PRESENTATION

2025

- Poster Presentation: Digestive Disease Week 2025, San Diego, CA, USA.
- Oral Presentation: Digestive Disease Week 2025, San Diego, CA, USA.

2024

- Oral Presentation: Researchers' Day 2024, Department of Medicine, Emory University, Atlanta, GA, USA.
- Poster Presentation: Digestive Disease Week 2024, Washington, D.C., USA.

2023

- Oral Presentation: Digestive Disease Research Day 2023, Atlanta, GA, USA.
- Poster Presentation: Researchers' Day 2023, Department of Medicine, Emory University, Atlanta, GA, USA.
- Oral Presentation: Digestive Disease Week 2023, Chicago, IL, USA.

2022

- Poster Presentation: Researchers' Day 2022, Department of Medicine, Emory University, Atlanta, GA, USA.
- Oral Presentation: Digestive Disease Research Day 2022, Atlanta, GA, USA.
- Poster Presentation: Digestive Disease Week 2022, San Diego, CA, USA.
- Oral Presentation: Digestive Disease Week 2022, San Diego, CA, USA.

2021

- Oral Presentation: Researchers' Day 2021, Department of Medicine, Emory University, Atlanta, GA, USA.

2019

- Oral Presentation: Probiotics & Prebiotics Congress: ASIA 2019, Taipei, Taiwan.

2018

- Poster Presentation: Life Science Annual Meeting 2018, Taoyuan, Taiwan.